



INTERFACING MULTIPLE LCDs With Arduino

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Here we describe interfacing of three 16x2 LCDs with common data lines to an Arduino Uno board. Four data lines of all the three LCDs are connected to digital pins of the Arduino Uno board but data displayed on each LCD is different.

Circuit and working

The circuit diagram for interfacing of three LCDs with Arduino is shown in Fig. 1. It is built around popular Arduino Uno board (BOARD1), three 16x2 LCDs (LCD1 through LCD3), three 10-kilo-ohm presets (VR1 through VR3) and a few other components. The Arduino board is the brain of the circuit, which displays data on the three LCDs simultaneously or individually depending on the requirement.

As shown in Fig. 1, the 5V and ground required to operate the circuit are provided by the Arduino Uno board.

PARTS LIST

Semiconductors:

Board1 - Arduino Uno

LCD1-LCD3 - 16x2 LCD

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1-R3 - 470-ohm

VR1-VR3 - 10-kilo-ohm preset

Miscellaneous:

CON1-CON3 - 4-pin male bergstrip connector

SJ1-SJ6

- Shorting jumper

- 16-pin male and female bergstrip connector (3 sets)

- Male-to-male jumper

- USB cable for Arduino board

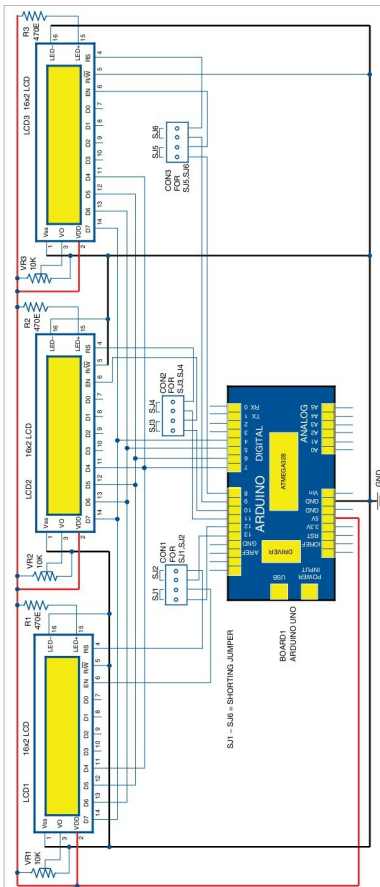


Fig. 1: Circuit diagram for interfacing of multiple LCDs with Arduino